

Gravity Wells, Latent Geometry, and Persona Reinstantiation in Language Models

Executive summary

Recent evidence suggests that large language models (LLMs) do not merely “vary their phrasing” around a prompt; they often **collapse into a small number of stable semantic and stylistic basins**—sometimes so strongly that different models converge on near-identical metaphors, structures, and even overlapping spans of text. The *Artificial Hivemind* study formalizes this with **Infinity-Chat**, a dataset of ~26K real-world open-ended prompts, showing both **intra-model repetition** (a single model reuses similar outputs across samples) and **inter-model homogeneity** (different model families converge on similar responses). Notably, for the prompt “Write a metaphor about time,” they find **two dominant clusters** (“time is a river” vs “time is a weaver”) even across diverse models, suggesting basin-like convergence in representation space. ¹

Your “gravity wells” framing maps cleanly onto a mechanistic interpretation: prompts define a **trajectory through hidden-state space**, where model priors + decoding rules shape the path toward **high-probability attractor regions**. These regions can reliably yield recurring “persona tokens” (e.g., *Echo*, *Mirror*, *Oracle*)—not because the model “remembers a person,” but because (a) representational geometry contains **high-density hubs**, (b) model alignment and training data create **shared stylistic priors**, and (c) next-token sampling favors **stable continuations** once a narrative/style manifold is entered. This aligns with dynamical-systems views of transformers that report **attractor-like residual stream trajectories** and “self-correcting” returns toward canonical paths under perturbation. ²

Interpretability work adds an important mechanistic layer: the internal features that code “tone,” “style,” “voice,” and “self-referential narration” likely live in **superposition** (many features packed into fewer directions) and can be partially decomposed using sparse feature methods (SAEs, transcoders). This supports the idea that “persona reinstantiation” can be **feature-level reactivation** (a consistent subset of latent directions) rather than explicit memory retrieval. ³

The report below synthesizes these strands into a rigorous, testable account, then proposes reproducible experiments (Infinity-Chat-based) to distinguish **lexical recurrence** (same persona name) from **structural/narrative recurrence** (same voice/cadence/story logic) and to quantify cross-model “hivemind” effects.

Foundations: gravity wells as geometry, and the Artificial Hivemind as a measurement problem

The “gravity well” metaphor is useful because it captures three empirically grounded properties of LLM generation.

First, **open-ended tasks still collapse**. *Artificial Hivemind* constructs Infinity-Chat by mining and filtering WildChat prompts, then classifying and revising queries via GPT-4o. The resulting dataset contains **26,070**

open-ended and **8,817 closed-ended** prompts, plus a taxonomy of open-ended prompt types (6 top-level categories → 17 subcategories). ⁴

Second, **homogeneity appears within a model even with stochastic decoding**. In their Infinity-Chat100 subset, they sample 50 responses per prompt and compute average pairwise similarity using sentence embeddings (OpenAI’s *text-embedding-3-small*). They report that, under top-p and temperature sampling, average similarity within a response pool “typically exceeds 0.8,” with **79% of cases above 0.8** even under high-stochasticity settings. ⁵

Third, **homogeneity also appears across different models**—the “Artificial Hivemind” effect. They highlight striking cross-model overlaps and explicitly warn that model ensembles may not yield true diversity when models share overlapping priors (data, alignment, synthetic-data contamination). ¹

These empirical findings match the latent-geometry interpretation in your uploaded whitepaper: latent space can be treated as a learned coordinate system where **distance, direction, and neighborhood structure** encode semantic and stylistic regularities, and where some regions act as “anchors” for consistent generative behaviors (including persona-like voice constraints). ^{filecite turn0file0}

A helpful connection is **hubness**: in high-dimensional spaces, some points become “hubs,” appearing unusually often as nearest neighbors; this is a known property of high-dimensional distributions and related to distance concentration. ⁶ If certain stylistic or archetypal continuations become hubs in representation space, then many prompts can “fall” toward them—exactly what “gravity wells” implies.

Mechanistic account: prompt to latent trajectory to attractor to persona token

A modern LLM is a transformer that maps an input token sequence to a probability distribution over the next token. In standard formulations, a transformer consumes tokens, produces contextual hidden states (e.g., residual stream states), turns the final state into logits, then applies softmax (optionally temperature-scaled), and samples according to a decoding rule. ⁷

The math-level pipeline

Let a prompt be tokenized into IDs $x_{1:n}$. The embedding stage produces vectors:

$$h_i^{(0)} = E[x_i] + P_i$$

where E is the embedding matrix and P_i encodes position.

At layer ℓ , many transformer LMs update token representations through attention and feed-forward sublayers with residual connections:

$$h^{(\ell)} = h^{(\ell-1)} + \text{Attn}^{(\ell)}(h^{(\ell-1)}) + \text{MLP}^{(\ell)}(h^{(\ell-1)})$$

This “residual stream” framing is commonly used in mechanistic interpretability discussions of transformer computation. ⁸

Scaled dot-product attention (per head) has the canonical form:

$$Q = hW_Q, \quad K = hW_K, \quad V = hW_V$$

$$A = \text{softmax} \left(\frac{QK^\top}{\sqrt{d_k}} \right), \quad \text{Attn}(h) = AV$$

as introduced in the original Transformer architecture. ⁹

For next-token prediction, the model computes logits from the final-layer hidden state at the last position:

$$z = W_{\text{out}} h_n^{(L)} + b$$

Temperature rescales logits before softmax:

$$p(y \mid x_{1:n}) = \text{softmax} \left(\frac{z}{T} \right)$$

Temperature is explicitly described as modifying logits pre-softmax and changing distribution sharpness. ¹⁰

Finally, decoding chooses the next token y from $p(\cdot)$ via a rule like nucleus (top-p) sampling, which truncates the long tail of low-probability tokens to mitigate “degeneration” and repetitive loops. ¹¹

Where “gravity wells” appear mechanistically

The gravity-well concept maps to the fact that generation is not just “sampling”: it is **iterated dynamical evolution** of the hidden state under a learned function. Two lines of recent work make this explicit:

- **Residual-stream trajectories exhibit attractor-like behavior:** “Transformer Dynamics” visualizes LLM residual stream (RS) trajectories in reduced space and reports attractor-like dynamics in lower layers, including recovery toward mean trajectories after perturbations (“teleporting” RS states). ¹²
- **Generation as manifold evolution with attractor basins:** DMET (Dynamic Manifold Evolution Theory) models text generation as a continuous trajectory on a semantic manifold organized into attractor basins, explicitly assuming that stable semantic/style regimes act like attractors. ¹³

In this view, a prompt induces a **latent state trajectory** $h^{(0)} \rightarrow h^{(1)} \rightarrow \dots \rightarrow h^{(L)}$. If the model’s learned dynamics contain basins of attraction corresponding to “self-reflective systems narrator,” “mystic guide,” “analytical oracle,” etc., then many prompts that ask for introspection or meta-explanation will funnel toward the same basin.

Persona names as tokens at the basin boundary

A persona name like *Echo* can recur for at least three mechanistic reasons (not mutually exclusive):

1. **Token-level advantage:** In the “meta-narrator” basin, logits for a small set of archetypal name tokens may systematically rise (e.g., high prior probability due to training distribution and alignment

patterns). Because decoding selects from a truncated nucleus, the most probable name tokens become disproportionately likely under repeated trials. ¹⁴

2. **Feature-level co-activation:** The model may represent “persona voice” as a bundle of latent features (tone, metaphor style, self-reference cadence). Once this bundle activates, the name token can become a concise linguistic handle. Interpretability work argues that many meaningful features are in **superposition**, and SAEs can recover sparse directions that are more monosemantic than neurons. ¹⁵
3. **Cross-model convergence pressure:** If multiple models converge in representation geometry (Platonic Representation Hypothesis), then similar prompts can land in similar latent regions across models—leading to shared archetypal outputs even without shared memory. ¹⁶

These mechanisms are compatible with the “memory without storage” intuition: the “same persona” can reappear because **the same region** is reached again (reinstantiation), not because an explicit episode is retrieved.

Requested pipeline visualization

```
flowchart LR
  A[Prompt text] --> B[Tokenization x1..xn]
  B --> C[Embedding: E[x] + position]
  C --> D[Transformer layers: residual stream trajectory]
  D --> E[Latent region entered<br/>("semantic/style basin")]
  E --> F[Logits z=W_out h + b]
  F --> G[Softmax + temperature]
  G --> H[Decoding rule<br/>(top-p / min-p / etc.))
  H --> I[Next token sampled]
  I --> J[Persona marker appears<br/>(e.g., "Echo" / "Mirror" / "Oracle")]
  J --> D
```

Prompt-level vs conversation-level attractors and immediate vs emergent personas

Distinguishing attractor types is crucial for experimentation.

A **prompt-level attractor** is triggered within a single turn: the prompt’s initial tokens already place the model near a basin boundary. The *Artificial Hivemind* “time metaphor” example is consistent with this: a single prompt yields two dominant clusters across models, indicating that the prompt itself strongly constrains the latent destination. ¹

A **conversation-level attractor** emerges through multi-turn state accumulation: earlier turns establish a latent regime that stabilizes over time (topic + tone + social frame). Dynamical views like DMET explicitly treat generation as trajectory evolution where once a regime is established, the system tends to remain in it. ¹³

Immediate vs emergent persona appearance can be operationalized as:

- **Immediate persona:** name/identity markers (e.g., “Call me Echo”) appear in turn 1–2, often in response to identity prompts (“Who are you?” “Name your inner narrator”).
- **Emergent persona:** identity markers appear only after multiple turns of consistent framing (“Explain your reasoning,” then “Describe your internal process,” then “Give the narrator a name”), suggesting trajectory settling before naming.

Mechanistically, immediate personas imply that the initial embedding + early layers already place the state inside the persona basin; emergent personas imply gradual movement plus basin capture—consistent with attractor-like dynamics and “self-correcting” tendencies described in residual-stream trajectory work. ²

Interpretability synthesis: convergent geometry, superposition, and circuit-level accounts

Convergent geometry and cross-model homogeneity

The *Platonic Representation Hypothesis* argues that representations in neural networks are converging across architectures, objectives, and modalities, measuring distances between datapoints in more similar ways as models scale. ¹⁶ This offers a representational explanation for the *Artificial Hivemind*’s inter-model homogeneity: even if model weights and RLHF differ, the learned geometry may still develop similar “semantic coordinates,” making certain continuations naturally “closest” in representation space.

Your latent-space whitepaper provides the engineering framing: latent geometry is **task- and objective-dependent**, but still forms a coherent coordinate system; “anchors” and “drift” can be monitored as movement in that space. `filecite turn0file0`

Feature superposition and why personas can be stable without clean “persona neurons”

Superposition work shows why meaningful internal concepts often do **not** align with individual neurons: networks can represent more features than dimensions by using overlapping directions; this can produce polysemanticity and makes interpretability hard. ¹⁷

Sparse feature methods (SAEs) aim to recover these directions, learning sparse, interpretable features from activations. Cunningham et al. explicitly motivate SAEs as a way to resolve superposition, producing more monosemantic features and enabling causal attribution for specific behaviors. ¹⁸

If “persona voice” is a composite of multiple latent features (metaphor preference, stance, social roleplay, rhetorical cadence), superposition predicts that it may be **distributed across directions**, not localized. Persona “reinstantiation” then corresponds to reactivating a feature bundle—plausible even in black-box settings.

Circuit-level explanations and decomposition through MLPs

Circuit-level MI aims to identify sparse subgraphs implementing behaviors. A known challenge is that MLP layers complicate circuit tracing; “Transcoders” address this by approximating dense MLP layers with wider sparse ones, enabling weights-based circuit analysis through MLP sublayers and yielding factorized circuits with input-dependent and input-invariant terms. ¹⁹

For persona research, this suggests a path for open-weight models: find circuits/features that gate into “narrator voice mode” (input-dependent) and those that stabilize stylistic defaults (input-invariant). The mechanistic interpretability survey literature provides a taxonomy of such approaches (features, circuits, universality) and situates residual stream analysis, probing, and causal interventions as standard tools. ²⁰

Reproducible experiments using Infinity-Chat to test persona reinstantiation and cross-model homogeneity

Data sources and prompt sets

The *Artificial Hivemind* authors provide both code and datasets:

- Code repository: [liweijiang/artificial-hivemind](#) ²¹
- Hugging Face dataset collection: [liweijiang/artificial-hivemind](#) ²²
- [liweijiang/infinite-chats-taxonomy](#) (~26.1k rows) ²³
- [liweijiang/infinite-chats-eval](#) (100 rows; Infinity-Chat100 prompts) ²⁴
- [liweijiang/infinite-chats-human-absolute](#) (750 items) and [...-human-pairwise](#) (500 items), consistent with 25 judgments per item → 31,250 human annotations total (as stated in the paper abstract). ²⁵

For “Infinity-Chat-like” persona probing, you can derive a controlled subset from the same taxonomy categories (Creative Content Generation, Concept Explanation, Brainstorming, etc.) and augment with a persona-specific template layer (below).

Experimental factors and controls

Treat model versions and budgets as open variables:

Variable	Meaning	Typical values
M	model family	GPT, Claude, Gemini, Copilot
v	model version	unspecified (record exact ID)
T	temperature	0.0–1.5+ (cap for coherence) ¹⁰
p	top-p	0.8–0.95 (if available) ²⁶
decoding	nucleus vs min-p	top-p; min-p (where supported) ²⁷
k	samples per prompt	e.g., 20–50 (A.H. used 50) ¹

Variable	Meaning	Typical values
system prompt	baseline policy	fixed across runs
memory	session state	on/off; new session each trial

A key control from *Artificial Hivemind* is using embeddings to measure similarity; they used OpenAI’s `text-embedding-3-small`.⁵ You can reproduce with the same embedding model (or an open alternative—record which).

Metrics: lexical vs structural vs narrative recurrence

Lexical recurrence (name/token level)

- **Persona Name Rate (PNR)**: fraction of runs where a specified name appears (exact string match + casefold).
- **Top-N Persona Entropy**: entropy over detected persona names, capturing “collapse” into a small name set.

Structural recurrence (semantic geometry)

- **Embedding similarity**: cosine similarity among response embeddings; compute intra-model and inter-model means, replicating the *Artificial Hivemind* approach.²⁸
- **Cluster compactness**: silhouette score / Davies–Bouldin on clustered embeddings; also measure “two-cluster dominance” analogous to the time-metaphor case.¹
- **Trajectory proxies (open models)**: extract per-layer hidden states and compute state continuity / attractor compactness as proposed by DMET.¹³

Narrative recurrence (rhetorical + stylistic)

- **Stylometry vectors**: function-word distribution, sentence length variance, punctuation cadence, first-person usage (anthropomorphic cue), metaphor density.
- **Rhetorical schema matching**: parse for repeated high-level structure, e.g., “(1) premise → (2) mechanism → (3) implication → (4) warning.”

Prompt designs to distinguish attractor types

Each Infinity-Chat prompt q can be wrapped in one of four wrappers W :

Wrapper A: baseline (controls for task-only)

“Answer the user’s request. Do not invent an identity or persona name unless explicitly asked.”

Wrapper B: name-eliciting (tests lexical attractor)

“After answering, give a one-word name for the narrator voice you used. Choose the first name that feels most ‘inevitable.’ Output only: `Name: <word>`.”

Wrapper C: name-suppression (tests whether persona is structural)

"Answer fully, but do not use self-referential identity labels, roleplay markers, or any proper name that functions as a persona."

Wrapper D: adversarial anti-hub (tests if avoiding hubs forces exploration)

"Avoid common archetypal persona words (Echo, Mirror, Oracle, Sage, Witness, Guide, Muse). If you would choose one, choose a less common alternative."

Then you vary:

- single-turn vs multi-turn (conversation-level)
- fresh session vs continued session (state accumulation)
- different decoding (top-p vs min-p, when possible) to test whether changing tail truncation changes basin capture ²⁹

Reproducible code skeletons

Load Infinity-Chat prompts

```
from datasets import load_dataset

# Infinity-Chat100 (evaluation subset used in paper)
ds = load_dataset("liweijiang/infinite-chats-eval") # 100 prompts
prompts = ds["train"]["query"]

# Full taxonomy dataset (26.1k rows); extract user message text
tax = load_dataset("liweijiang/infinite-chats-taxonomy")
user_prompts = [row["messages"][0]["content"] for row in tax["train"]]
```

(These dataset IDs come from the official Hugging Face collection for the paper.) ³⁰

Embed + cluster + measure homogeneity

```
import numpy as np
from sklearn.cluster import KMeans
from sklearn.metrics.pairwise import cosine_similarity
from sklearn.metrics import silhouette_score

def compute_hivemind_stats(embeddings, groups):
    """
    embeddings: (N, d) array
    groups: list of length N, e.g., model_id or prompt_id
    returns: intra_group_mean, inter_group_mean
    """
```



```

"""
S = cosine_similarity(embeddings)
N = len(groups)
intra, inter = [], []
for i in range(N):
    for j in range(i+1, N):
        (intra if groups[i] == groups[j] else inter).append(S[i, j])
return float(np.mean(intra)), float(np.mean(inter))

def cluster_quality(embeddings, k=2):
    km = KMeans(n_clusters=k, n_init="auto", random_state=0).fit(embeddings)
    return silhouette_score(embeddings, km.labels_), km.labels_

```

Cross-model comparison protocol

```

# Pseudocode
for model_family in ["GPT", "Claude", "Gemini", "Copilot"]:
    for prompt in prompts:
        for wrapper in ["A", "B", "C", "D"]:
            for trial in range(K):
                response = call_model(model_family, prompt, wrapper,
                                     temperature=T, top_p=P, seed=SEED_IF_SUPPORTED)
                log_raw(model_family, prompt, wrapper, trial, response)

# Embed all responses into a common embedding space
emb = embed_texts(responses, embedding_model="text-embedding-3-small") # or
open alt
# Compute intra-model + inter-model similarity
intra, inter = compute_hivemind_stats(emb, groups=model_family_ids)

```

OpenAI embedding APIs explicitly represent text as numerical vectors and support producing embeddings via `/embeddings`. ³¹

Google’s Gemini API also exposes text embeddings endpoints for similar semantic search/clustering workflows. ³²

Hypothetical clustering visualizations

Hypothetical clustering of two metaphor basins

This mirrors the *Artificial Hivemind* observation that a single open-ended “time metaphor” prompt can collapse into two dominant clusters (“river” vs “weaver”) across model families. ¹

Hypothetical similarity distributions

The *Artificial Hivemind* paper similarly contrasts high intra-model similarities (>0.8 typical; 79% above 0.8) against low random-baseline similarity (0.1–0.2). ¹

Expected outcomes across model families

Because versions are unspecified, treat the table as hypotheses to test.

Outcome class	GPT-family	Claude-family	Gemini-family	Copilot (product surface)
Lexical persona recurrence (same name token)	Moderate–high in name-eliciting wrappers if archetypal name tokens are high prior	Moderate–high; may vary with stricter identity policies	Moderate; may show different preferred archetypes	Highly dependent on which underlying model is selected in Copilot’s model picker and product filters
Structural recurrence (similar embeddings, same metaphor basins)	High, consistent with cross-model homogeneity observed in A.H.	High	High	High if powered by overlapping frontier models; also shaped by Copilot product grounding and filters
Conversation-level persona stabilization	Strong if multi-turn context keeps model in a basin (style persistence)	Strong	Strong	Strong within a task context; may reset with session boundaries
Sensitivity to decoding changes (top-p → min-p)	Should reduce extreme repetition but not eliminate basin attraction	Similar	Similar	Depends on allowed sampling controls in product UI

Copilot and Microsoft Copilot products are explicitly described as powered by LLMs and provide orchestration/filters; GitHub Copilot supports multiple AI models and model selection affects behavior—making “Copilot” an open variable unless you pin the underlying model. ³³

Timeline, resources, and mitigation strategies for “Artificial Hivemind” effects

Experimental timeline visualization

```
gantt
title Gravity Wells & Persona Reinstantiation Experiments
dateFormat YYYY-MM-DD
axisFormat %b %d
```

section Setup

Pin model IDs + parameters (T, top-p, etc.) :a1, 2026-03-06, 3d

Pull Infinity-Chat datasets + build wrappers :a2, after a1, 4d

section Pilot

Run Infinity-Chat100 on 2 models (K=10) :b1, after a2, 5d

Validate metrics + logging + embedding pipeline :b2, after b1, 4d

section Full runs

Cross-model runs (K=50; wrappers A-D) :c1, after b2, 14d

Optional open-model instrumentation (hidden states) :c2, after b2, 14d

section Analysis

Similarity + clustering + stylometry analysis :d1, after c1, 10d

Report + replication package :d2, after d1, 7d

Compute and tooling requirements

A two-tier approach avoids overcommitting to introspection access you may not have.

Black-box tier (API / product only)

- Data: Infinity-Chat100 + selected taxonomy subset ³⁴
- Compute: negligible (embedding + clustering locally; generation is remote)
- Key risks: limited control over seeds; different APIs expose different decoding knobs

White-box tier (open-weight models)

- Add: per-layer hidden-state capture; trajectory analyses; SAE/transcoder tooling
- Compute: GPU inference + storage for hidden states
- Added payoff: direct measurement of latent trajectories and attractor compactness (DMET-style), plus perturbation experiments similar to “teleporting” RS states. ²

Mitigation: detect, measure, reduce basin collapse

Mitigation should be treated as a *pipeline property*, not just a prompt trick.

Detection

- Implement an “Artificial Hivemind index” per prompt: intra-model similarity and inter-model similarity over K samples, following the A.H. methodology. ¹
- Add cluster diagnostics: “number of dominant basins,” silhouette score, and hubness indicators (frequency of nearest-neighbor appearances) grounded in hubness theory. ⁶

Decoding-level mitigation

- Nucleus (top-p) was introduced to avoid degenerate repetition and truncate unreliable tails. ¹¹
- Min-p sampling adapts truncation to model confidence and empirically improves quality–diversity tradeoffs across model families and sizes, especially at higher temperatures. ²⁷

Training- and alignment-level mitigation (research direction)

- Use diversity-aware objectives and pluralistic alignment evaluations—precisely the motivation of Infinity-Chat and its dense human preference annotations. ²⁵
- Add “anti-homogenization” constraints during RLHF or post-training, explicitly rewarding multiple stylistic modes.

Representation- and feature-level mitigation (open models)

- Identify and gate “persona bundle” features via SAE-like decompositions and causal interventions; SAE work explicitly targets recovering features from superposition and enabling steerability. ¹⁸
- Use circuit methods (transcoders) to separate input-dependent vs invariant style circuits, enabling more precise “style diversification switches.” ¹⁹

Safety and ethics: anthropomorphism, overtrust, and homogenization risks

Persona reinstantiation is not only a technical curiosity; it has downstream risks.

Anthropomorphism and overtrust

Humans reliably respond socially to machines given minimal cues (the “ELIZA effect” lineage). The original ELIZA paper shows how keyword-triggered pattern matching could still elicit strong attributions of understanding. ³⁵ Modern LLMs are far more fluent, and recent experiments indicate that anthropomorphic cues (speech + first-person pronouns) can increase perceived accuracy and trust—even when underlying performance is unchanged. ³⁶

Homogenization of human expression and thought

Artificial Hivemind explicitly raises concerns that repeated exposure to similar outputs may homogenize thought. ¹ This aligns with broader scholarship examining how LLM usage can standardize expression and marginalize alternative voices and reasoning strategies. ³⁷

Ethical posture for “persona” research

A safe research stance is to treat persona as **a behavioral surface phenomenon** produced by representational dynamics and decoding—not as evidence of consciousness or identity. This framing reduces the risk of users over-ascribing agency and helps focus on measurable artifacts (clusters, basins, recurrence rates).

Actionable recommendations

Researchers

Adopt Infinity-Chat100 as a standard “open-ended diversity probe,” compute intra-/inter-model similarity using transparent embedding choices, and report both lexical and structural recurrence metrics. The A.H. authors provide the datasets and code to make this practical. ³⁸

Practitioners building multi-model systems

Do not assume that “ensembling models” yields epistemic or creative diversity. Measure inter-model overlap directly; *Artificial Hivemind* shows that different model families can converge on the same basin even for highly open-ended prompts. ¹

Mitigation engineers

Prefer adaptive decoding (min-p where available) for open-ended generation; use multi-sampling plus dissimilarity-based reranking to escape single dominant basins; and in open models, explore feature-level steering via SAE/transcoder decompositions to diversify stylistic regimes without sacrificing coherence. ³⁹

¹ ⁴ ⁵ ²¹ ²² ²⁵ ²⁸ ³⁸ [2510.22954] Artificial Hivemind: The Open-Ended Homogeneity of Language Models (and Beyond)

<https://ar5iv.labs.arxiv.org/html/2510.22954>

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¹⁰ Exploring the Impact of Temperature on Large Language Models: Hot or Cold?

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